Fiscal Year:	FY 2013	**Task Last Updated:**	FY 06/14/2013
PI Name:	Almeida, Eduardo A C Ph.D.		
Project Title:	The Role of the p53 Pathway in Spaceflight-Induced Tissue Degeneration		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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City:	Moffett Field	**State:**	CA
Zip Code:	94035	**Congressional District:**	18
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2007 Space Flight: Bion-M1 NNH07ZTT001N
Start Date:	08/01/2009	**End Date:**	09/30/2014
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Smith, Jeffrey	**Contact Phone:**	650-604-0880
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Flight Program:	Bion/Shuttle		
Flight Assignment:	STS-135 ; Bion M1 NOTE: Extended to 9/30/2014 per A. Chu/ARC (Ed., 4/3/2014) NOTE: Extended to 7/31/2014 per A. Chu/ARC (Ed., 4/30/2013)		
Key Personnel Changes/Previous PI:	N/A		
COI Name (Institution):	Globus, Ruth (NASA Ames Research Center)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Our working hypothesis is that spaceflight factors (including radiation and microgravity) affect the growth and regenerative health of a broad range of mammalian tissues, including bone, muscle, and immune system, by causing indiscriminate slowing or arrest of normal cell proliferation. Specifically we hypothesize that there is a key regulatory cell-signaling pathway responsible for this space-induced arrest, namely the p53/p21 signaling pathway, and that either radiation and/or microgravity can activate it. We have already tested this hypothesis with spaceflight tissues from STS-131, validating some of its key aspects such as p21activation in osteoblasts, and are now utilizing mouse tissues from the recently completed Bion M1 30-day spaceflight to test the p53 hypothesis with long-term spaceflight tissue samples. Specifically we are working towards determining if the p53/p21 pathway is also altered during 30-day spaceflight in bone tissues by: 1) By measuring gene expression and protein expression and modifications of key signaling molecules that control cell cycle arrest and proliferation such as those in the p53/p21 and PI3Kinase pathways 2) Immunohistochemically identifying specific cell types in bone with active p53/p21 3) Correlating p53/p21 pathway and related cell cycle status to bone loss measurements by histomorphometry, microCT and transmission R-ray microscopy.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The significance of the studies already completed and still underway is several-fold, but first and foremost they are a the test of an original and novel hypothesis that has the potential to be a common element of the already known tissue degenerative conditions observed in space, and an important predictor of yet unknown degenerative conditions that may emerge in long-term spaceflight, specially outside of LEO. There is already incontrovertible ground evidence that the p53 signaling pathway is key in the response to radiation, and there is also compelling evidence that p53 is required for tissue degenerative conditions caused by mechanical unloading. Although spaceflight evidence is limited, skin and muscle in rats flown in space also show elevated p53, strongly compelling us to follow this line of inquiry. The fact that p53 pathway is a powerful tumor suppressor, and inducer of cell cycle arrest (through p21) and programmed cell death, or apoptosis (through p21), should serve as clear indicator if spaceflight factors activate the p53/p21 pathway, this may have systemic effects on all the body processes involving cell proliferation. These processes are most notably the indispensable tissue regeneration that keep every organ from collapsing due to unrepaired cellular and tissue damage. In addition these studies are also highly significant because they attempt to pinpoint the cellular specificity of the p53/p21 pathway activation in bone, and to determine if this is a organ-wide process or limited to certain cell types such as the bone marrow osteoprecursor somatic stem cell lineage. If the p53/p21 pathway is proven to be significantly involved in spaceflight arrest of tissue regeneration and degenerative conditions, it will offer both a new pharmacological target for intervention by drugs such as pifithrin class inhibitors and as a biomarker of tissue damage in space. Finally the Bion M1 studies we are conducting will shed new light on the old question of bone loss in space, possibly extending our understanding of bone dynamics in adult animals, and by being the longest experimental rodent exposure to spaceflight to date. Already our recently published results on bone dynamics during STS-131 show that both osteoclasts, osteoblasts, and osteocytes are involved in bone loss in response to spaceflight, and that both p21and p53 are altered. When the ongoing Bion M1 tissue and cell studies are concluded we expect that we will have gained new factual understanding of bone loss in space under conditions previously not attempted, and will have an important new perspective about the potential for activation of the p53/p21 pathway to be a serious problem for human tissue regeneration during long-term spaceflight, and in human health and disease related to disuse.
Task Progress:	April 2023 Update Added a Life Sciences Space Research publication to the Bibliography for this project as NASA funding is acknowledged (Ed., 4/6/23) After significant delays the the 30-day Bion M1 Russian orbital spaceflight mission was conducted between April and May 2013, and 6 flight C57B6 male mice and 7 vivarium controls were made available to the NASA biospecimen sharing program principal investigators for tissue recovery and analysis. The major accomplishment for the year was the successful recovery of flight and vivarium mouse tissues and live bone marrow stem cell cultures just concluded June 11. The time leading towards spaceflight was also used in detailed preparation for the science flight activities in Moscow, as well as to analyze and publish previous results from the BSP shuttle flights. Specifically during the grant year, final science procedures were prepared and tested in 3 experiment verification tests (EVTs) in which mice were dissected, as in the flight, samples prepared for microCT, histology, biochemical, and genomic analyses, and live bone marrow cultures differentiated in-vitro into mineralized bone nodules. In addition to conducting EVTs, our preparations included logistical aspects of flight such as reagent procurement and placement in Russia as well as shipping of supplies and equipment to set up laboratory operations at the Institute for Biomedical Problems of the Russian Academy of Sciences (for dissection and initiation of post-flight cell cultures) and at the Koltsov Developmental Biology Institute (KDBI) of the Russian Academy of Sciences. In addition to the science verification work described above we concluded the analysis of bone samples from the 15-day STS-131 spaceflight and published a detailed and comprehensive accounting of the effects of spaceflight and microgravity on the p53/p21 cell cycle arrest pathway, as well as accompanying effects (Microgravity Induces Pelvic Bone Loss through Osteoclastic Activity, Osteocytic Osteolysis, and Osteoblastic Cell Cycle Inhibition by CDKN1a/p21.PLOS ONE, April 2013, Volume 8, Issue 4, e61372). In brief we have now determined that short-term spaceflight (15 days) induces significant bone loss in the pelvic-femoral body-weight load-center of C57B6 mice, and that the multiple cellular and tissue mechanisms of bone loss are activated, including ostoclast activation, osteocytic osteolysis activation and osteoblastic cell cycle arrest mediator induction. Specifically we found that the p53/p21 signaling pathway was altered by spaceflight with marked increases in the expression of CDKN1a/p21 in osteoblasts, and generalized suppression of p53. These results validate the hypothesis that cell cycle arrest in osteoprogenitors is activated by spaceflight through p21, but also shows that cells are not induced to undergo p53 mediated programmed cell death/apoptosis, as indicated by decreased p53 levels and unaltered cell death. Going forward we are currently preparing to conduct August 2013 Moscow ground controls in the spaceflight BLOC hardware, as well as repeating the "bridge" vivarium controls common to both flight and ground asynchronous controls.
Bibliography Type:	Description: (Last Updated: 12/08/2025)

Abstracts for Journals and Proceedings	Blaber E, Dvorochkin N, Burns BP, Almeida EAC. "Microgravity Suppresses Global Pro-Differentiation Gene Expression Patterns in Mouse Bone Marrow Derived Stem Cells." 28th Annual Meeting of the American Society for Gravitational and Space Research, New Orleans, LA, November 27-December 2, 2012. 28th Annual Meeting of the American Society for Gravitational and Space Research, New Orleans, LA, November 27-December 2, 2012. Program and abstracts, p. 94. https://www.asgsr.org/images/stories/pdf/2012_AnnualMeeting_Program.pdf , Nov-2012
Abstracts for Journals and Proceedings	Blaber E, Dvorochkin N, Burns BP, Almeida EAC. "Gene Expression Analysis of Mouse Bone and Bone Marrow Exhibit Significantly Altered Expression of Cell Proliferation Signaling Pathways in Response to Spaceflight." EMBL 14th Annual PhD Symposium, Heidelberg, Germany, October 25-27, 2012. EMBL 14th Annual PhD Symposium, Heidelberg, Germany, October 25-27, 2012. , Oct-2012
Abstracts for Journals and Proceedings	Blaber E, Dvorochkin N, Vercoutere W, Yousuf R, Alwood J, Burns BP, Almeida EAC. "Microgravity alters matrix-integrin–kinase signaling cascades resulting in arrest of the cell cycle in bone osteoprogenitor cells." IAC 63: 63rd International Astronautical Congress, Naples, Italy, October 1-5, 2012. IAC 63: 63rd International Astronautical Congress, Naples, Italy, October 1-5, 2012. , Oct-2012
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Articles in Peer-reviewed Journals	Radugina EA, Almeida EAC, Blaber E, Poplinskaya VA, Markitantova YV, Grigoryan EN. "Exposure to microgravity for 30 days onboard Bion M1 caused muscle atrophy and decreased regeneration in murine femoral quadriceps." Life Sci Space Res (Amst). 2018 Feb;(16):18-25. https://doi.org/10.1016/j.lssr.2017.08.005 ; PMID: 29475516 , Feb-2018
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